

VARIABLE ANNUITIES: A UNIFYING VALUATION APPROACH

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ABSTRACT. Life annuities and pension products usually involve a number of guarantees, such as minimum accumulation rates, minimum annual payments or a minimum total payout. Packaging different types of guarantees is the feature of so-called variable annuities. Basically, these products are unit-linked investment policies providing a post-retirement income. The guarantees, commonly referred to as GMxBs (namely, Guaranteed Minimum Benefits of type 'x'), include minimum benefits both in case of death and survival. In this paper we propose a unifying framework for the valuation of variable annuities under quite general model assumptions. We compute and compare contract values and fair fee rates under 'static' and 'mixed' valuation approaches, via ordinary and least squares Monte Carlo methods, respectively.

1. INTRODUCTION

The term variable annuity is used to refer to a wide range of life insurance products, whose benefits can be protected against investment and mortality risks by selecting one or more guarantees out of a broad set of possible arrangements. A variable annuity is a long-term, tax-deferred investment, designed for obtaining a post-retirement income. Several rider benefits may be underwritten for the period before retirement, in particular death benefits and minimum accumulation returns. Indeed, a variable annuity can be shaped so as to offer dynamic investment opportunities with some guarantees, protection in case of early death and/or a post-retirement income.

Key words and phrases. Variable annuities, post-retirement income, risk management, guarantees, Least Square Monte Carlo.

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In respect of traditional life insurance products, the main features of variable annuities are represented by a large variety of possible guarantees, which can be underwritten either for the accumulation, the annuity or the death benefit, as well as tax incentives. Guarantees are briefly referred to as GMxB, Guaranteed Minimum Benefit of type 'x', where 'x' stands for accumulation (A), death (D), income (I) or withdrawal (W). All GMxB's provide a protection of the policyholder's savings account: the GMAB during the "accumulation phase" (i.e., the working period of the policyholder); the GMDB in case of early death, during the accumulation phase and possibly for some years after retirement; the GMIB and the GMWB after retirement, in particular in face of high longevity. Basically, a variable annuity is a fund-linked insurance contract, including a package of financial options on the policy account value (see Smith (1982) and Walden (1985)). Guarantees are then also looked at as riders to the basic benefit given by the account value. A description of the main characteristics of variable annuity products, and the development of their market as well, is provided by Ledlie *et al.* (2008).

In some countries, the success of variable annuities is also due to tax incentives, introduced by governments to support the development of individual pension solutions and contain public expenditure. Typically, tax reductions are granted during accumulation, so that a variable annuity can be considered a tax-deferred investment. Annuity payments are taxable as, or more beneficially than, ordinary income; tax penalties may be applied in case funds are withdrawn before a given age. For more details on variable annuities taxation see, for example, Brown and Poterba (2006).

The design of variable annuities matches features of unit-linked life insurance contracts (namely, the 'linking' mechanism) to those of participating contracts (the guarantees). Contrarily to most participating policies, guarantees in variable annuities are explicit and a fee is applied to meet their cost. While insurers are reluctant to continue to provide guarantees whose cost is not charged to the policyholder, significant advances in hedging techniques enable them to offer more sophisticated guarantees. However, available financial models are not directly suitable for the relevant pricing and hedging, as options embedded in insurance products imply an interaction between financial and mortality/longevity risks; moreover, in some situations, namely when the guarantee concerns the choice of the post-retirement income, policyholder preferences significantly affect the value of the guarantee.

As can be easily understood, many risks affect the variable annuity business; mortality/longevity, financial, policyholder behaviour risks are the most relevant. It is not fully known how these risks interact, especially when referring to the post-retirement phase. An appropriate approach to the risk management of variable annuity liabilities is then required. Some preliminary investigations have already been reported. Sun (2006) examines the risks associated with various combinations of guaranteed benefits; Gilbert *et al.* (2007) describe the findings of a survey on the risk management of guarantees within variable annuity products in the US. Risk management issues are also addressed by Ledlie *et al.* (2008).

Among the different phases of the risk management process, the pricing and hedging of guarantees, i.e. of the relevant financial options, should be

a major concern for the insurer when designing the contract. Appropriate evaluation techniques need to be developed in order to account on one hand for the interaction between financial and mortality/longevity issues, on the other for the policyholder behaviour.

Possible approaches to the pricing and hedging of the most common guarantees in variable annuity arrangements have already been discussed in the literature. Recently, special attention has been addressed to withdrawal benefits, thanks to the fact that, when appropriately managed, they can provide a flexible post-retirement income. With regard to the pricing of this class of benefits, Milevsky and Salisbury (2006) assess their cost and compare their findings with prices quoted in the market; Dai *et al.* (2008) develop a singular stochastic control model, and investigate the optimal withdrawal strategy for a rational policyholder; Chen and Forsyth (2008) describe an impulse stochastic control formulation, while Chen *et al.* (2008) explore the effect of alternative policyholder behaviours. In all such contributions, most of the attention is addressed to financial risks. Mortality is considered, e.g., by Milevsky and Posner (2001), dealing with the pricing of guaranteed minimum death benefits. The valuation of a variable annuity combining death and withdrawal guarantees is the target of Bélanger *et al.* (2009), while a general framework for the pricing of guarantees under the assumption of optimal policyholder behaviour is suggested by Bauer *et al.* (2008). The case of lifelong withdrawals is dealt with by Holz *et al.* (2007) and Piscopo (2009).

The purpose of this paper is to provide a unifying framework for the valuation of variable annuities. In Section 2, taking a risk management perspective, we outline the main risks that affect the cost of guarantees, stressing in particular the need for an integrated approach. In Section 3 we describe the most common guarantees in variable annuity products and refer to contributions available in the literature (in particular, to Ledlie *et al.* (2008)) as well as to insurers' informative material for further details. Since it is burdensome to provide an exhaustive representation of all products available in the market, we choose to focus mainly on those offered in Europe. Then, in Section 4 we deal with the pricing of guarantees under three different assumptions on the policyholder behaviour, referred to as the static, mixed and dynamic approaches. To this end, we adopt the comprehensive model developed by Bacinello *et al.* (2010a). Section 5 contains extensive numerical examples under the static and mixed approaches. The examples show the flexibility of the valuation framework and explore the sensitivity of fair fee rates and contract values to underlying assumptions. Some final remarks in Section 6 conclude the paper.

2. THE RISK MANAGEMENT PERSPECTIVE

Several risks affect the performance of a life insurance portfolio: the traditional risk of random fluctuations in the number of deaths (namely the 'volatility' or *process risk*), the market risk, the expense risk, the credit risk, the policyholder behaviour risk and so on. Long-term insurance products, for example life annuities, are also exposed to the risk of systematic deviations in the number of deaths, i.e. the *uncertainty risk*, due to the randomness in level and trend of future mortality. The impact of both mortality/longevity

and investment risks can be rather high in the presence of some types of guarantees and options, as it happens, in particular, in variable annuity products.

A Risk Management (RM) process usually requires several phases. In the first, the *risk identification*, each source of risk should be investigated by singling out its components, in particular hedgeable and unhedgeable, and other relevant factors such as the portfolio size, the prevailing type of benefit, and so on.

The *risk assessment* (the second step of the RM process) should allow for the interaction among several risks. When dealing with life annuities, in particular, an integrated approach to mortality and financial risks, as well as policyholder behaviour risks, is required. Usually, the complexity of the assessment problem implies the adoption of Monte Carlo simulation techniques.

The choice of (a mix of) risk management techniques, or *RM actions*, constitutes the third step of the RM process. Risk control actions involve the choice of the life table, the structure of profit participation, and the features and pricing of minimum guarantees, as well as internal risk reduction actions (diversification strategies). Conversely, risk financing actions consist of risk transfers, either via traditional reinsurance arrangements, or capital market solutions. In the following we do not deal with the latter issues, but mainly attack the pricing of minimum guarantees, allowing for several risk sources that affect their value. We note that risks related to some specific guarantees can be shared between the insurer and the policyholder. For example, the annuitisation rate, namely the rate at which the amount resulting from an accumulation process is converted into a life annuity, can be stated at the beginning of the accumulation period rather than at its end or sometime between. Depending on this choice, the relevant risk is differently shared between the insurer and the policyholder, with obvious consequences on the cost of the guarantee. Further, it is worthwhile mentioning that diversification strategies combining opposite benefits with respect to the lifetime ('natural hedging' techniques), naturally arise within variable annuities, as emerges in Section 3. Packaging survival and death benefits under the same insurance policy (hedging 'across time'), as it occurs in variable annuities, originates a partial offsetting between longevity and mortality risk. Indeed, if mortality rates decrease, then life annuity costs increase while death benefit costs decrease, and vice versa.

For a more detailed discussion on the RM perspective we refer to Pitacco *et al.* (2009) and Bacinello *et al.* (2010b).

3. THE STRUCTURE OF VARIABLE ANNUITIES

In this section we illustrate what kind of benefits can be provided by a variable annuity and introduce the relevant notation.

3.1. Benefits, assets, premiums, expenses. As mentioned in Section 1, variable annuities merge the most attractive commercial features of unit-linked and participating life insurance contracts: dynamic investment opportunities, protection against financial risks, benefits in case of early death. Further, they offer modern solutions in regard of the post-retirement income,

trying to arrange a satisfactory trade-off between annuitisation needs and bequest preferences. Basically, a variable annuity is a fund-linked insurance contract, with rider benefits in the form of guarantees on the policy account value; such riders become attainable either in case of death or survival.

Available guarantees are referred to as GMxB, where ‘x’ stands for the class of benefits involved. They may be first grouped into two main broad classes:

- Guaranteed Minimum Death Benefits (GMDB);
- Guaranteed Minimum Living Benefits (GMLB).

The second class can be further arranged into three subclasses:

- Guaranteed Minimum Accumulation Benefits (GMAB);
- Guaranteed Minimum Income Benefits (GMIB);
- Guaranteed Minimum Withdrawal Benefits (GMWB).

Variable annuities are generally issued with a single premium or single recurrent premiums. The total amount of premiums is also named the principal of the contract or the invested amount. Apart from some upfront costs, premiums are entirely invested into the reference funds chosen by the policyholder. Several investment opportunities are available to the customer, providing different risk-return profiles. Thus, the policyholder can opt for more conservative or more dynamic asset combinations. She is also allowed to switch from one risk-return solution to another at no cost, if some constraints are fulfilled (for example, the switch is required no more than once a year). Unlike the with-profit or participating business, reference funds backing variable annuities are not required to replicate the guarantees selected by the policyholder, as these are hedged by specific assets. Therefore, reference fund managers have more flexibility in catching investment opportunities.

Guarantees and asset management fees, administrative costs and other expenses are charged year by year to the contract through a reduction of the policy account value. This improves the transparency of the contract, as any deduction to the policy account value must be reported to the policyholder. Some guarantees can be added or removed, at policyholder discretion, when the contract is already in-force. Accordingly, the corresponding fees start or stop being charged. The cost of guarantees, as well as other expenses, are typically expressed as a given percentage of the policy account value.

3.2. Benefits in variable annuities. In this section we provide some details on the contents of each GMxB. We will mainly refer to contracts available in the European market, but the framework can be easily adapted to policies sold in other countries.

We consider a single premium variable annuity contract issued at time 0. Let P denote the single premium and $T \geq 0$ the end of the accumulation period; if $T = 0$ the annuity is immediate and hence some guarantees become meaningless. We only consider single premium contracts as they constitute the building block for single recurrent premium schemes. We denote by A_t the policy account value at time t . Clearly, this value depends on the evolution of the reference fund in which the single premium is invested. We assume that all possible guarantees are selected at time 0 and are kept for

all the contract duration. To simplify notation, we assume that no partial withdrawal occurs prior to retirement time.

The GUARANTEED MINIMUM DEATH BENEFIT (GMDB) is usually available during the accumulation period; some insurers are willing to provide a GMDB also after retirement, up to some maximum age (say, 75 years). The structure of the guarantee is as follows: in case of death prior to the stated maturity, the insurer will pay the greater between the account value and a guaranteed amount. Thus, in case of death at time t , during the accumulation period (i.e. $t \leq T$), the benefit is given by

$$b_t^D = \max\{A_t, G_t^D\}, \quad t \leq T, \quad (3.1)$$

where G_t^D is some guaranteed amount. The guaranteed amount can be either fixed or depending on the account value. Examples of a fixed guaranteed amount are as follows.

- Amount of premiums paid, net of partial withdrawals (the so-called return of premiums). Under our assumptions (single premium and no partial withdrawals):

$$G_t^D = P. \quad (3.2)$$

- The roll-up of premiums at a specified guaranteed interest rate:

$$G_t^D = Pe^{\delta t} \quad \delta > 0, \quad (3.3)$$

where δ is the guaranteed instantaneous interest rate. Guarantee (3.3) could be modified to include a time-dependent, deterministic guaranteed interest rate; this would allow us to model situations in which the roll-up feature applies only until a given age of the insured and/or the rate is adjusted at regular intervals. We mention that if the policyholder withdraws funds from her account during the accumulation period, the guaranteed amount G_t^D is reduced accordingly.

Examples of a guaranteed amount depending on the account value are as follows.

- The highest account value recorded at some specified times prior to death (ratchet guarantee). Under our assumptions (single premium and no partial withdrawals):

$$G_t^D = \max_{t_i < t} A_{t_i}. \quad (3.4)$$

As a consequence of the guarantee, the profits of the reference fund are locked-in at the (predetermined) ratchet dates t_i , $i = 0, 1, \dots, n$ and $0 = t_0 < t_1 < \dots < t_n < T$.

- The account value at some prior specified date, known as reset date (reset guarantee):

$$G_t^D = A_{\max\{t_i: t_i < t\}}, \quad (3.5)$$

where times t_i , $i = 1, 2, \dots, n$, now define the reset dates. The difference between the ratchet and the reset guarantee stands in the behaviour of the guaranteed minimum amount: in the ratchet guarantee the minimum amount never decreases, whilst a reduction may occur in the reset, if the account value decreases between two reset dates.

Obviously, (3.4) and (3.5) should be adjusted in order to account for premiums or withdrawals occurring after the ratchet/reset dates.

Further specifications of the guaranteed amount, e.g. including cliquet guarantees, could be considered. If the contract does not include a GMDB, the death benefit is simply given by A_t ; it can still be described through (3.1) by setting

$$G_t^D = 0. \quad (3.6)$$

Combinations of guarantees are also possible; for example

$$G_t^D = \max \left\{ Pe^{\delta t}, \max_{t_i < t} A_{t_i} \right\} \quad (3.7)$$

formalizes a combination of roll-up and ratchet features.

Similarly to the GMDB, the GUARANTEED MINIMUM ACCUMULATION BENEFIT (GMAB) is available prior to retirement. At some specified date (typically, the end of the accumulation period), the insured, if alive, is credited the greater between the policy account value and a guaranteed amount G_T^A :

$$b_T^A = \max\{A_T, G_T^A\}, \quad (3.8)$$

where b_T^A is the benefit. As before, the guaranteed amount may be given by

- the amount of premiums paid, net of partial withdrawals;
- the roll-up of premiums, net of partial withdrawals, at a specified guaranteed interest rate;
- the highest account value recorded at some specified times (ratchet).

Thus, G_T^A is expressed by formulae resembling (3.2)–(3.7). A further guarantee which may be attached to the GMAB is the reset, which gives the opportunity to renew the GMAB when it reaches maturity. We stress then that the term *reset* has a different meaning within a GMAB and a GMDB: a reset option in the GMAB simply allows to postpone the maturity date T .

The GUARANTEED MINIMUM INCOME BENEFIT (GMIB) provides a life-time annuity from a specified future point in time. The guarantee may be arranged in two different ways.

- The amount to be annuitised will be the greater between the account value and a specified sum, G_T^I . Possible ways to specify such an amount are similar to the GMAB; then, G_T^I is defined by formulae resembling (3.2)–(3.7). The annuitisation rate, η , will be defined according to market conditions prevailing at the annuitisation date. Then, the periodic (annual, monthly, ...) flat life annuity payment is given by

$$b^I = \eta \max\{A_T, G_T^I\}. \quad (3.9)$$

- The annuitisation rate will be the more favourable between a stated rate g and η . The annuitised amount will be the account value. Then, the periodic payment of the life annuity is expressed as

$$b^I = A_T \max\{\eta, g\}. \quad (3.10)$$

This case is usually referred to as the Guaranteed Annuity Option (GAO). It is implicit in (3.10) the assumption that the GAO is exercised if it is in the money, neglecting subjective preferences regarding annuitisation vs. bequest, as well as asymmetric information.

As noted in Section 2, different sharings between the insurer and the policyholder of the risks emerging after retirement (in particular the longevity risk) are implied by (3.9) and (3.10). The former guarantee is sometimes described as a guarantee on the annual amount, which would suggest an arrangement similar to a deferred life annuity; it is then worthwhile to stress that the guarantee actually concerns the amount to be annuitised, as described above. After annuitisation, the policyholder loses access to the account value (while prior to annuitisation the contract works like an investment product, bearing some guarantees). The guarantee must be selected by the policyholder some years before annuitisation; typically, the GMIB may be exercised after a waiting period of 5 to 10 years. The cost of the GMIB is subtracted from the account value during the accumulation period. If prior to annuitisation the policyholder gives up the guarantee, the insurer stops deducting the relevant fee. Typically, full annuitisation is required; however, partial annuitisation is admitted in some arrangements. As far as the duration of the annuity is concerned, the following solutions are usually available: a traditional life annuity; a last-survivor annuity; a life annuity with a minimum number of payments (say, up to 5 or 10 years). Money-back or capital protection arrangements may also be available, providing a death benefit consisting of the residual principal amount, i.e. the annuitised amount net of the annual payments already cashed. The annual amount may be either fixed, participating or indexed to inflation or stock prices; a financial risk is borne by the annuitant in the latter case, as the annual amount can fluctuate in time (conversely, in a participating scheme the annual amount never decreases). Payments frequency can vary according to policy conditions.

In respect of duration, in the following we just consider three cases, that are the most common in practice:

- (Ia) a whole life annuity;
- (Ib) an annuity-certain with maturity $T' > T$, in which payments occur independently of life contingencies;
- (Ic) an annuity-certain with maturity $T' > T$, followed by a deferred whole-life annuity if the insured is alive at T' .

Moreover, we consider neither money-back arrangements nor last-survivor benefits, and assume flat annuity payments, as in (3.9) and (3.10).

It is worthwhile to mention that several variants can be found for the GMIB in respect of the structure described above. For example, the policyholder could choose the annuity option exercise date, or there could be a guaranteed annuity payment until the last annuitisation time, even if the account is exhausted. However, for the sake of brevity, we do not address these variants.

The **GUARANTEED MINIMUM WITHDRAWAL BENEFIT** (GMWB) allows periodical withdrawals from the policy account, even if the account value reduces to zero either because of bad investment performances or a long lifetime of the insured. The guarantee concerns the periodic payment and the duration of the income stream. The periodic payment is stated as a given percentage, β_t , of a base amount, W_t , which for example can be given by the account value when the guarantee is first selected by the policyholder (since we have assumed that all guarantees are selected at time 0, this would

coincide with the single premium P). We denote by

$$b_t^W = \beta_t W_t$$

the contractually specified withdrawal amount, where t belongs to a set of fixed withdrawal dates. In some arrangements, at specified dates (e.g., every policy anniversary) the base amount may step up to the current value of the policy account, if this is higher; this is a ratchet guarantee, which may be lifetime or limited to a given period (e.g., 10 years). Note that, thanks to the ratchet, the guaranteed payment may increase in time; in some arrangements, a maximum accepted annual increase is stated in policy conditions. The guaranteed payment can alternatively be the exact, the maximum or the minimum amount that the policyholder is allowed to withdraw. In the latter case, any withdrawal above the guaranteed level reduces the base amount. The duration of the withdrawals may be fixed (e.g., 20 years) or lifetime. In the former case, if at maturity the account value is positive, it is paid back to the policyholder or, alternatively, the contract stays in-force until exhaustion of the policy account value. We assume that withdrawals can occur

- (Wa) up to a date $T' > T$, independently of survival;
- (Wb) up to a date $T' > T$, only if the insured is alive;
- (Wc) during the whole life of the insured.

The cost of the guarantee is deducted from the account value during the withdrawal period; if the policyholder gives up the guarantee, the relevant fee stops being applied. During this period, the policyholder keeps access to the unit-linked fund; if at the time of death the account value is still positive, it is paid to the policyholder estate.

From the descriptions above, it emerges that the GMAB and GMDB are similar to what can be found in investment life insurance contracts, apart from the possible range of guarantees, which is wider in variable annuities than in traditional contracts. The GMIB is like a traditional life annuity, possibly participating. The GMWB is the real novelty of variable annuities in respect of traditional life insurance contracts; it provides a benefit which is similar to an income drawdown, but with guarantees. When comparing a GMIB to a GMWB, three major differences arise: the duration of the annuity (which is usually lifetime in the GMIB), the accessibility to the account value (just for the GMWB) and the features of the reference fund (which is unit-linked in the GMWB, but typically participating in the GMIB). Clearly, the presence of death benefits also in the GMIB, a lifetime duration for the withdrawals in the GMWB and other possible features reduce a lot the differences between the GMIB and the GMWB.

4. VALUATION FRAMEWORK

In this section we assess the cost of guarantees under quite general model assumptions. We show how all the features of variable annuity benefits, as described in Section 3, can be encompassed into the general framework proposed by Bacinello *et al.* (2010a). In particular, we will consider three alternative assumptions concerning the policyholder behaviour, that lead to corresponding valuation approaches, called *static*, *dynamic* and *mixed*.

4.1. Preliminaries. It is a well-known result in asset pricing theory (see, e.g., Duffie (2001)) that, under reasonable economic assumptions, the market price of a security is given by its expected discounted cash-flows. Discounting is performed at the risk-free rate and the expectation is taken with respect to a suitably risk-adjusted probability measure. The incompleteness of insurance markets implies that infinitely many such probabilities exist. We assume henceforth that the insurer has picked out a specific probability for valuation purposes. From now on, without further mention, all random variables and processes will be considered under this probability. Expectation is denoted by $E[\cdot]$, while $E_t[\cdot]$ is the expectation conditional on information available at time t . We let r_t be the instantaneous risk-free rate at time t and

$$M_t = e^{\int_0^t r_u du}, \quad t \geq 0,$$

is the balance at time t of a money market account formalizing the investment of cash at the instantaneous risk-free rate r . Finally, we denote by S_t the unit value at time t of the reference fund backing the contract, and by τ the time of insured's death.

We now formalize the cash-flows generated by the contract. To this end, we recall that b_t^D is the lump sum benefit paid in case of death at time t and denote by B_t^L the cumulated benefits paid in case of survival up to time t .

4.2. The static approach. The so-called *passive*, or *static*, valuation approach (see, e.g., Milevsky and Salisbury (2006)) is characterized by the following assumptions:

- the policyholder does not withdraw any fund from her account during the accumulation period or during the payout phase of a GMIB rider;
- if the contract contains a GMWB rider, the policyholder withdraws exactly the amounts contractually specified;
- the contract is never surrendered.

4.2.1. Cumulated survival benefits. Assume that the contract provides only one of the following alternative living benefits: (i) a benefit paid at time T (GMAB); (ii) a deferred annuity from time T onwards (GMIB); (iii) a guaranteed minimum withdrawal benefit (GMWB) from time T onwards.

Let us now specify the cumulated survival benefits.

- (i) GMAB: the benefit paid at T is expressed by (3.8), so that the cumulated survival benefit is given by

$$\begin{aligned} B_t^L &= \begin{cases} 0 & \text{if } t < T \\ b_T^A & \text{if } t \geq T \end{cases} \\ &= b_T^A 1_{t \geq T}, \end{aligned}$$

where 1_E denotes the *indicator* of the event E , which takes value 1 if E is true and 0 otherwise.

- (ii) GMIB.

(Ia) *Whole-life annuity*:

$$B_t^L = b^I \sum_{i=1}^{\infty} 1_{t \geq T_i},$$

with annuity payment dates $(T \leq) T_1 < T_2 < \dots$

(Ib) *Annuity certain*:

$$B_t^L = b^L \sum_{i=1}^m 1_{t \geq T_i},$$

with annuity payment dates $(T \leq) T_1 < T_2 < \dots < T_m (\leq T')$.

(Ic) *Annuity certain followed by a deferred whole-life annuity*: We take B_t^L exactly as in case (Ia).

In (Ia), (Ib) and (Ic) the annuity payment b^L can be expressed, e.g., by formulae (3.9) or (3.10). In Section 4.2.2 we will define a ‘fictitious’ death benefit for (Ib) and (Ic).

(iii) GMWB.

(Wa) *Temporary withdrawals independent of survival*:

$$B_t^L = \sum_{i=1}^m b_{T_i}^W 1_{t \geq T_i} + \max \{A_{T'}, 0\} 1_{t \geq T'}, \quad (4.1)$$

where $(T \leq) T_1 < T_2 < \dots < T_m (\leq T')$ are the withdrawal dates. The second term on the right-hand side of (4.1) represents the payment at date T' , if the insured is alive, of the remaining policy account value (thus, we disregard policy solutions which allow for a continuation of the contract beyond time T' , until fund exhaustion). Again, in Section 4.2.2 a ‘fictitious’ death benefit will be defined.

(Wb) *Temporary withdrawals in case of survival*: We take B_t^L exactly as in case (Wa).

(Wc) *Whole life withdrawals*:

$$B_t^L = \sum_{i=1}^{\infty} b_{T_i}^W 1_{t \geq T_i}, \quad (4.2)$$

where $(T \leq) T_1 < T_2 < \dots$ are the withdrawal dates.

4.2.2. *Death benefit*. Let us now define the benefit b_t^D paid in case of death at time t . If $t \leq T$, it is usually given by (3.1). Moreover, if there is a GMIB rider of type (Ib) and (Ic), in the static approach it is perfectly equivalent, from the valuation point of view, to consider as death benefit the value at time t of the annuity payments due between t and T' when $T < t \leq T'$. If instead there is a GMWB rider, we have

$$b_t^D = \begin{cases} \max \{A_t, R_t\} & \text{for } T < t \leq T' \text{ in case (Wa)} \\ \max \{A_t, 0\} & \text{for } T < t \leq T' \text{ in case (Wb)} \\ \max \{A_t, 0\} & \text{for } t > T \text{ in case (Wc)} \end{cases}.$$

Here R_t denotes the value at time t of the remaining contractually guaranteed withdrawal amounts due between t and T' . Then, in case of death before T' , we have implicitly assumed that withdrawals cease and the market value of the remaining guaranteed amounts is immediately paid as a death benefit. This solution, however, is not financially equivalent to the withdrawals continuation until T' due to the presence of the proportional fees that reduce the account value while the contract is still in-force. Finally, we

set $b_t^D = 0$ when $t > T$ for GMAB and GMIB (Ia), when $t > T'$ for GMIB (Ib,c) and GMWB (Wa,b).

4.2.3. *Valuation.* Denote by φ the proportional fee rate that is applied to the account value in order to recover the cost of all the guarantees. In this static approach, the instantaneous evolution of the account value during the life of the insured can be formally described as follows:

$$dA_t = \begin{cases} A_t \frac{dS_t}{S_t} - \varphi A_t dt - dB_t^L & \text{if } A_t > 0 \\ 0 & \text{otherwise} \end{cases}, \quad (4.3)$$

and $A_0 = P$. Then, the return on the account value is that of the reference fund, adjusted for fees and survival benefits. Denoting by B_t the (total) cumulated benefits paid by the contract up to time t , we have

$$B_t = (B_{\tau^-}^L + b_\tau^D) 1_{\tau \leq t} + B_t^L 1_{\tau > t}, \quad t \geq 0. \quad (4.4)$$

Finally, the value at time t of the variable annuity policy is given by its expected discounted cash-flows

$$V_t = E_t \left[\int_t^\infty \frac{M_t}{M_u} dB_u \right], \quad t \geq 0. \quad (4.5)$$

In some situations this value can be expressed in closed-form, once a very natural assumption of stochastic independence between financial and demographic risk factors is made. This is the case of the celebrated single premium arrangement analysed by Brennan and Schwartz (1976) and Boyle and Schwartz (1977), where the contract contains both a GMDB and a GMAB rider. However, if more sophisticated assumptions do not allow to obtain such closed-form formulae, a straightforward application of the Monte Carlo method can be carried out in order to value the expectation in (4.5).

Since V_t is net of fees, (4.5) gives the value of the variable annuity contract as a function of the fee rate φ . Of course the contract is fairly priced if and only if

$$V_0 \equiv V_0(\varphi) = P, \quad (4.6)$$

so that a fair fee rate φ^* is implicitly defined as a solution of (4.6).

4.3. **The dynamic approach.** Unlike the static approach, in the so-called *active* or *dynamic* approach the policyholder is assumed to withdraw amounts not necessarily coinciding with those contractually specified under a GMWB rider; in particular she can decide not to withdraw, or to surrender the contract. In addition, partial withdrawals or surrender decisions could be made on dates not coinciding with those contractually specified, as well as during the accumulation period. Sometimes these actions are admitted also in the payout phase of a GMIB, although this could introduce some adverse selection risk. In almost all of these cases there are penalties applied to the withdrawn amounts. That is, in case of surrender, the policyholder could receive less than the account value, while in case of partial withdrawal the amount received, net of penalties, could be less than the amount subtracted from the account value. Moreover, guaranteed amounts, if present, are accordingly reduced.

Once a strategy of partial (or total) withdrawals is given, the corresponding process of cumulated survival benefits is obtained by suitably modifying

that defined in Section 4.2.1. By a withdrawal strategy we mean a stochastic process specifying the amount to withdraw at each date (provided the insured is alive), including possibly the surrender value. Clearly, only withdrawal strategies consistent with contractual constraints are *admitted*. We need now to consider two cumulated survival benefits processes, one including all net withdrawal amounts, the other including instead gross withdrawal amounts. Denoting by θ a specific withdrawal strategy, we let $(NB)_t^{L,\theta}$ and $(GB)_t^{L,\theta}$ be the corresponding net and gross cumulated survival benefits, and add the superscript θ to all relevant symbols previously introduced. We replace equation (4.3) with

$$dA_t^\theta = \begin{cases} A_t^\theta \frac{dS_t}{S_t} - \varphi A_t^\theta dt - d(GB)_t^{L,\theta} & \text{if } A_t^\theta > 0 \\ 0 & \text{otherwise} \end{cases}, \quad (4.7)$$

and $A_0^\theta = P$. If the withdrawal strategy θ implies the surrender of the contract, we denote by λ^θ the corresponding surrender time; conventionally, we set $\lambda^\theta = \tau$ if the surrender is not implied by the withdrawal strategy. Given θ , the contract value at time t is expressed as

$$V_t^\theta = E_t \left[\int_t^\infty \frac{M_t}{M_u} dB_u^\theta \right], \quad t \geq 0, \quad (4.8)$$

but now with

$$\begin{aligned} B_t^\theta &= \left((NB)_{\tau^-}^{L,\theta} + b_\tau^{D,\theta} \right) 1_{\tau \leq \min\{t, \lambda^\theta\}} \\ &\quad + (NB)_t^{L,\theta} 1_{t < \min\{\tau, \lambda^\theta\}} + (NB)_{\lambda^\theta}^{L,\theta} 1_{\lambda^\theta < \tau, \lambda^\theta \leq t}, \quad t \geq 0. \end{aligned} \quad (4.9)$$

Finally, the contract value is

$$V_0 = \sup_{\theta} V_0^\theta, \quad (4.10)$$

where the supremum is taken with respect to all admissible withdrawal strategies. Equation (4.10) gives the value of the contract under the assumption of an ‘optimal’ policyholder behaviour (corresponding to the worst case scenario for the insurer). Note, however, as pointed out by Milevsky and Salisbury (2006), that this behaviour is optimal from the insurer’s point of view, but may not necessarily be the best for the policyholder, who could have different preferences (leading thus to a different choice of the risk-adjusted probability measure) and/or asymmetric information, see also Bacinello *et al.* (2010a).

In this paper we do not offer specific results following this fully dynamic approach, which of course requires a numerical scheme. Some results concerning a contract with a GMWB rider are presented, e.g., by Milevsky and Salisbury (2006), Dai *et al.* (2008), Chen *et al.* (2008) and Chen and Forsyth (2008).

4.4. The mixed approach. In this section we assume that the policyholder is ‘semiaactive’, in the sense that, in presence of a GMWB rider, she withdraws exactly when and what contractually specified, and does not make any partial withdrawal during the accumulation period or during the payout phase of a GMIB. However, unlike the static approach, at any time during the life of the contract the policyholder may decide to surrender.

We denote by b_t^S the surrender value, net of any penalty, that the policyholder receives in case of surrender at time $t > 0$. If surrender is not admitted for some t , e.g. during the payout phase of a GMIB rider in order to rule out any adverse selection from the portfolio of life annuitants, then $b_t^S = 0$.

Unlike the dynamic approach, we isolate now the surrender benefit from the other (contractually specified) withdrawal amounts, so that we do not need to modify the cumulated survival benefits and the death benefit defined in Section 4.2 or to distinguish between net and gross survival benefits. Hence, (4.3) still holds; however, the surrender value must be added to the total cumulated benefits defined in (4.4). To this end, let λ denote the surrender time, conventionally equal to τ if the policyholder never surrenders the contract. Given λ , the total cumulated benefits generated by the contract up to time t , denoted by B_t^λ , are now

$$B_t^\lambda = (B_{\tau-}^L + b_\tau^D) 1_{\tau \leq \min\{t, \lambda\}} + B_t^L 1_{t < \min\{\tau, \lambda\}} + (B_\lambda^L + b_\lambda^S) 1_{\lambda < \tau, \lambda \leq t}, \quad t \geq 0, \quad (4.11)$$

so that the corresponding value of the contract, denoted by V_t^λ , is

$$V_t^\lambda = E_t \left[\int_t^\infty \frac{M_t}{M_u} dB_u^\lambda \right], \quad t \geq 0. \quad (4.12)$$

Finally, the time 0 value of the contract (net of insurance fees) is given by the following expression:

$$V_0 = \sup_{\lambda} V_0^\lambda, \quad (4.13)$$

where the supremum is taken with respect to all possible surrender times. Once again, the fair fee rate φ^* is implicitly defined by equation (4.6).

The supremum in (4.13) needs to be evaluated by means of a numerical approach. Among the possible methods that have been proposed in the literature to solve such kind of problems (binomial or multinomial trees, partial differential equations with free boundaries, Least Squares Monte Carlo), we choose the Least Squares Monte Carlo method. Our problem fits perfectly in the general framework dealt with by Bacinello *et al.* (2010a), so that we refer to their paper for an accurate description of the philosophy underlying this method and the valuation algorithm. In Appendix A we provide a brief description of the Least Square Monte Carlo algorithm adapted to our framework.

4.5. Comparison. In order to relate the approaches described in Sections 4.2, 4.3 and 4.4, we denote by V_0^{static} , V_0^{dynamic} and V_0^{mixed} the contract values at time 0 defined respectively by (4.5), (4.10) and (4.13). By their very definition, it turns out

$$V_0^{\text{static}} \leq V_0^{\text{mixed}} \leq V_0^{\text{dynamic}}.$$

Indeed, from the insurer's point of view, the dynamic approach assumes the worst case scenario since the policyholder can choose among all withdrawal strategies and, in particular, the surrender time. In the mixed approach, instead, the policyholder can choose only the surrender time, so that her 'optimal' strategy is selected within a subset of that considered by a dynamic

agent. Finally, the static approach defines a single, specific, withdrawal strategy included in the previous subset. As a consequence, the proportional fees that have to be applied to the account value to make the contract fair are ordered in the same way: they are the highest with the dynamic approach and the smallest with the static one.

5. NUMERICAL INVESTIGATIONS

We consider the single premium variable annuity contract described in Sections 3 and 4 and evaluate different types of guarantees under the static and the mixed approach. For the former, we use a standard Monte Carlo procedure, while for the latter the Least Square Monte Carlo algorithm described in Appendix A. In particular, we determine the fair fee rate with the static approach (denoted by φ^*) and explore its sensitivity with respect to contract parameters such as, e.g.: minimum interest rate guaranteed for the roll-up of premium, frequency of ratchet dates, guaranteed annuity conversion rates and guaranteed total withdrawals. Moreover, we compare the contract values under the static and mixed approaches for different fee rates, surrender penalties, market and biometric parameters.

Table 1 synthesizes the guarantees analysed and the contract features that are kept fixed across all examples. Most examples consist of a GMDB, a single type of GMLB, and/or their combination. As far as GMIBs are concerned, we only consider the guaranteed annuity option (see (3.10)), since guarantees of the annuitised amount (as defined in (3.9)) are equivalent to GMABs.

	x	T	T'	static	static & mixed
Roll-up GMDB, GMAB, GMAB & GMDB	60	5	—	Table 3	Table 12
Ratchet GMDB, GMAB, GMAB & GMDB	60	5	—	Table 4	—
Roll-up GMDB, GMIB, GMIB & GMDB (Ia,b,c)	60	5	10 (Ib,c)	Tables 5, 6, 7	Table 13, 18, 19
Ratchet GMDB, GMIB, GMIB & GMDB (Ia,b,c)	60	5	10 (Ib,c)	Tables 8, 9, 10	—
GMWB (Wa)	—	0	5, 10, 15, 20	Table 11	Table 14, 15, 16

TABLE 1. Map of the numerical examples.

As can be seen from Table 1, in almost all the numerical examples we have fixed the entry age at $x = 60$ and the end of the accumulation period at $T = 5$. Indeed, those data are very common features of variable annuities sold in the Italian market. For the GMWB we have fixed $T = 0$, i.e. we have considered immediate withdrawals since proportional fees for GMWBs are applied during the withdrawal period. We have taken as base amount $W_t = P = A_0$ and assumed that the annual withdrawal rate is $\beta = \frac{1}{T'}$; in this way, at least the entire premium will be paid back to the policyholder

or her estate during the withdrawal period. Note, however, that the fair fee rates reported in Table 11 would be the same should we set $T > 0$ and if the base amount were reset to the account value at T , provided the current level of the state variables in T were the same as in 0.

We now briefly describe the financial and mortality model employed in the valuation. We do not deal with the calibration of this model to real market data, as our purpose is just to illustrate the flexibility of the valuation approach. We adopt a slightly simplified version of the framework presented in Bacinello *et al.* (2009); all processes are specified under the selected risk-adjusted probability measure. The instantaneous risk-free rate r and the variance K of the reference fund follow square root processes:

$$\begin{aligned} dr_t &= \xi_r(\zeta_r - r_t)dt + \sigma_r\sqrt{r_t}dZ_t^r, \\ dK_t &= \xi_K(\zeta_K - K_t)dt + \sigma_K\sqrt{K_t}dZ_t^K, \end{aligned}$$

while the log of the reference fund price S satisfies the following dynamics:

$$d \log S_t = \left(r_t - \frac{1}{2}K_t \right) dt + \sqrt{K_t} \left(\rho_{SK}dZ_t^K + \sqrt{1 - \rho_{SK}^2}dZ_t^S \right).$$

The force of mortality μ evolves according to the mean reverting square root process

$$d\mu_t = \xi_\mu(\hat{\mu}(t) - \mu_t)dt + \sigma_\mu\sqrt{\mu_t}dZ_t^\mu, \quad (5.1)$$

where $\hat{\mu}(t) = c_1^{-c_2}c_2(x+t)^{c_2-1}$ is the deterministic Weibull intensity. In the following we denote by $\hat{\tau}$ the residual lifetime having force of mortality $\hat{\mu}$. The standard Brownian motions Z^r , Z^K , Z^S and Z^μ are assumed to be independent. The residual lifetime τ is then linked to μ through

$$\tau = \inf \left\{ t \geq 0 : \int_0^t \mu_s ds > \mathcal{E} \right\},$$

where $\mathcal{E}$ is a unit exponential random variable independent of $(\mu_t)_{t \geq 0}$. The force of mortality μ_t drives the instantaneous probability of death at time t conditional on survival for an individual aged x at time 0. The unconditional survival probability is given by

$$P(\tau > t) = E \left[e^{-\int_0^t \mu_s ds} \right]$$

and can be computed in closed form given the affine nature of μ (see Biffis and Millosovich (2006)). According to (5.1), the force of mortality μ is pushed towards the moving target $\hat{\mu}$ and the dynamics is noised by the Brownian term, amplified by the size of μ .

The parameters underlying these processes are reported in Table 2. The

r	K	S	A	μ
$r_0 = 0.03$	$K_0 = 0.04$	$S_0 = 100$	$A_0 = 100$	$\mu_0 = \hat{\mu}(0)$
$\xi_r = 0.60$	$\xi_K = 1.50$	$\rho_{SK} = -0.70$		$\xi_\mu = 0.50$
$\zeta_r = 0.03$	$\zeta_K = 0.04$			$\sigma_\mu = 0.03$
$\sigma_r = 0.03$	$\sigma_K = 0.40$			$c_1 = 90.43$
				$c_2 = 10.36$

TABLE 2. Parameters used in the simulation.

coefficients c_1 and c_2 defining the Weibull intensity $\hat{\mu}$ were obtained by fitting the survival probabilities for a male aged 60 implied by the projected life table IPS55, commonly used in the Italian annuity market. In Figure 1 we sketch the survival and density function of τ and $\hat{\tau}$. Note that, for the chosen parameters, we have $E[\tau] = 28.30$ and $E[\hat{\tau}] = 26.61$ (see Table 17).

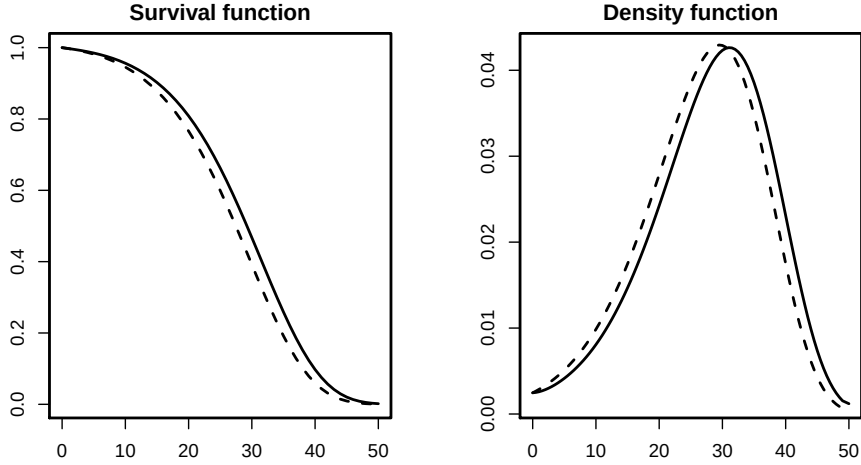


FIGURE 1. Survival and density function of τ and $\hat{\tau}$ (solid and dashed line, respectively)

Results using the static approach were obtained by averaging over 10 sets of 200 000 simulations each, those using the mixed approach by averaging over 10 sets of 20 000 simulations each. In the Least Square Monte Carlo algorithm we have employed 4 state variables (K , A , r and μ) and up to 4-th degree powers as basis functions (see Appendix A for more details). The market annuitisation rate η of GMIBs was computed in closed form exploiting the joint affine structure of r and μ (see Biffis and Millosovich (2006)).

In Tables 3-11 we report the fair fee rate φ^* obtained in the static approach for different types of guarantees and different values of the guaranteed roll-up rate δ , the ratchet frequency, the guaranteed annuitisation rate g and the guaranteed withdrawal rate β .

$\delta\%$	GMDB	GMAB	GMAB & GMDB
0	0.024	3.201	3.263
1	0.027	4.245	4.340
2	0.031	6.039	6.229
3	0.035	12.168	—
4	0.039	—	—

TABLE 3. $\varphi^*\%$ – roll-up – static approach.

First note that, as expected, higher guarantees imply higher contract values and in turn higher fair fees. By looking at the numerical results reported in Tables 3 to 10, it can be noticed that the GMDB is practically valueless. Indeed, the fair fee rate required for this guarantee ranges from 0.024% to

ratchet frequency	GMDB	GMAB	GMAB & GMDB
1/12	0.047	7.862	8.050
1	0.036	5.829	5.945
2	0.031	5.022	5.114
3	0.028	4.383	4.462
4	0.026	4.148	4.220

TABLE 4. φ^* % – ratchet – static approach.

$g\%$		≤ 5.00	5.25	5.50	5.75	6.00
GMIB		0.000	0.012	0.779	1.662	2.507
$\delta\%$	GMDB	GMIB & GMDB				
0	0.024	0.024	0.037	0.806	1.691	2.538
1	0.027	0.027	0.040	0.809	1.695	2.542
2	0.031	0.031	0.043	0.813	1.699	2.547
3	0.035	0.035	0.047	0.817	1.703	2.551
4	0.039	0.039	0.052	0.822	1.708	2.557

TABLE 5. φ^* % – roll-up, case (Ia) – static approach.

$g\%$		≤ 11.0	11.5	12.0	12.5	13.0
GMIB		0.000	0.175	1.015	1.827	2.605
$\delta\%$	GMDB	GMIB & GMDB				
0	0.024	0.024	0.201	1.043	1.856	2.637
1	0.027	0.027	0.204	1.047	1.860	2.641
2	0.031	0.031	0.207	1.050	1.864	2.645
3	0.035	0.035	0.212	1.055	1.868	2.650
4	0.039	0.039	0.216	1.059	1.874	2.655

TABLE 6. φ^* % – roll-up, case (Ib) – static approach.

$g\%$		≤ 5.00	5.25	5.50	5.75	6.00
GMIB		0.000	0.060	0.930	1.813	2.658
$\delta\%$	GMDB	GMIB & GMDB				
0	0.024	0.024	0.086	0.958	1.843	2.690
1	0.027	0.027	0.089	0.961	1.847	2.694
2	0.031	0.031	0.093	0.965	1.851	2.698
3	0.035	0.035	0.097	0.969	1.855	2.703
4	0.039	0.039	0.101	0.974	1.860	2.709

TABLE 7. φ^* % – roll-up, case (Ic) – static approach.

0.047%. This is because, given the length of the accumulation period and the policyholder age at inception, the GMDB is unlikely to become effective.

Another common feature of Tables 3 to 10 is that the fair fee rate for a contract including both a GMDB and a GMLB is greater than the sum of the fair fee rates for the corresponding guarantees taken separately. This result, although counterintuitive at first sight, can be explained as follows. It is doubtless that, with the static approach, the cost of both guarantees packaged together must be equal to the sum of the costs of the unbundled guarantees because we are using the same risk-adjusted pricing measure and,

	$g\%$	≤ 5.00	5.25	5.50	5.75	6.00
	GMIB	0.000	0.012	0.779	1.662	2.507
ratchet frequency	GMDB	GMIB & GMDB				
1/12	0.047	0.047	0.059	0.828	1.713	2.560
1	0.036	0.036	0.048	0.817	1.702	2.549
2	0.031	0.031	0.043	0.812	1.698	2.545
3	0.028	0.028	0.041	0.810	1.695	2.542
4	0.026	0.026	0.039	0.808	1.693	2.540

TABLE 8. $\varphi^*\%$ – ratchet, case (Ia) – static approach.

	$g\%$	≤ 11.0	11.5	12.0	12.5	13.0
	GMIB	0.000	0.175	1.015	1.827	2.605
ratchet frequency	GMDB	GMIB & GMDB				
1/12	0.047	0.047	0.223	1.066	1.878	2.659
1	0.036	0.036	0.212	1.055	1.867	2.648
2	0.031	0.031	0.207	1.050	1.863	2.643
3	0.028	0.028	0.205	1.047	1.860	2.640
4	0.026	0.026	0.203	1.045	1.858	2.639

TABLE 9. $\varphi^*\%$ – ratchet, case (Ib) – static approach.

	$g\%$	≤ 5.00	5.25	5.50	5.75	6.00
	GMIB	0.000	0.060	0.930	1.813	2.658
ratchet frequency	GMDB	GMIB & GMDB				
1/12	0.047	0.047	0.109	0.980	1.865	2.712
1	0.036	0.036	0.097	0.969	1.854	2.701
2	0.031	0.031	0.093	0.964	1.849	2.696
3	0.028	0.028	0.090	0.961	1.847	2.694
4	0.026	0.026	0.088	0.959	1.845	2.692

TABLE 10. $\varphi^*\%$ – ratchet, case (Ic) – static approach.

T'	5	10	15	20
$\beta\%$	20.00	10.00	6.67	5.00
GMWB	3.919	1.953	1.207	0.820

TABLE 11. $\varphi^*\%$ – case (Wa) – static approach.

in particular, the same mortality basis (even if this might be different in practice). However, this cost is not paid as a lump sum but it is recovered through a reduction of the account value according to a proportional fee rate. Then, for a given guarantee, the higher the fee rate applied to the account value the lower the base to which it will be applied in the future (i.e., the account value itself). To fix the ideas, denote by φ^D and φ^L the fair fee rates for GMDB and GMLB taken separately. If we apply the fee rate $\varphi^D + \varphi^L$, the account value will be lower than that resulting from the separate application of φ^D or φ^L , and hence the total charge for the guarantees recovered during the whole duration of the contract will be insufficient. This explains why a rate φ^{DL} higher than $\varphi^D + \varphi^L$ is required when both guarantees are sold together.

We note from Table 3 that in some instances no fee rate can make the contract fair. Indeed, if the guarantee is too high, the account value at time 0 may be insufficient to recover its cost. This is the case, e.g., of a GMAB with roll-up of premium and a 4% roll-up rate or of a GMAB & GMDB with a roll-up rate higher than 3%.

Some comments concerning GMIBs are in order. First of all, from Tables 5 to 10 one can see that there is a threshold for the guaranteed annuitisation rate under which the guarantee is likely to be out of the money at maturity (i.e., at the annuitisation date T). Secondly, comparing the results with those of Tables 3 and 4, one can notice that a guaranteed annuity option can be much cheaper than a guarantee on the annuitised amount. Recall in fact that the results reported in Tables 3 and 4 pertain also to GMIBs when the guarantee is on the annuitised amount. Just to fix the ideas, with a roll-up rate of 2% on the annuitised amount, a GMIB requires a fee of about 6% (see Table 3) while a GAO with annuitisation rate of 6% in case (Ia) requires a fee of about 2.5% (see Table 5).

As far as ratchet guarantees are concerned, note the sensitivity of the fair fee rate when the ratchet frequency decreases from monthly to annual. There is also a slight decrease in the fair fee rate when the frequency changes from 3 to 4 years, although in both cases there is only one ratchet date after time 0.

Concerning withdrawals, for brevity, we just consider the most common case in practice, i.e. (Wa), in which withdrawals occur independently of survival. From Table 11 one can notice that, as obvious, the proportional fee rate required in order to make the contract fair strongly decreases with the contract duration.

Coming now to the results obtained using the mixed approach, we analyse first the case of roll-up guarantees with combined GMDB and GMAB or GMIB (Ia), as well as GMWB (Wa), once again for brevity. More specifically, in Tables 12 to 14 we report the contract values at time 0 for given levels of the fee rate and the surrender penalty, in order to compare them with the corresponding results obtained using the static approach (displayed in the last column). To this end, we have assumed that the surrender value is given by $b_t^S = A_t(1 - p)$ for $t < T$, for GMAB & GMDB, where p denotes the surrender penalty. Moreover, for GMIB & GMDB we have assumed

$$b_t^S = \begin{cases} A_t(1 - p) & t < T \\ A_t & t = T \\ 0 & t > T \end{cases}.$$

Finally, with GMWB we have assumed, for $t < T'$, that $b_t^S = A_t(1 - p)$ if t is not a withdrawal date, and $b_t^S = b_t^W + \max\{A_t - b_t^W, 0\}(1 - p)$ otherwise.

From Tables 12-14 one can note that, in spite of the penalty, the contract values obtained with the mixed approach are almost always greater than those obtained with the static approach. Indeed, the presence of fees could make convenient an early termination of the contract. We have chosen fee rates near the fair fee rate computed with the static approach, but of course one expects that with lower rates the contract value V_0^{mixed} will approach V_0^{static} . When the contract value is greater than P , the fixed fee rate and/or

GMAB & GMDB

$\varphi\%$	$p\%$						V_0^{static}
	0	1	2	3	4	5	
6	103.557	103.056	102.618	102.189	101.788	101.488	100.399
7	102.654	102.182	101.726	101.299	100.882	100.548	99.123
8	101.843	101.380	100.928	100.515	100.136	99.787	98.088
9	101.122	100.674	100.237	99.848	99.479	99.130	97.270
10	100.461	100.037	99.629	99.251	98.898	98.564	96.644
11	99.875	99.474	99.090	98.723	98.388	98.083	96.177

TABLE 12. V_0^{mixed} – roll-up, $\delta^A = \delta^D = 2\%$.

GMIB & GMDB

$\varphi\%$	$p\%$						V_0^{static}
	0	1	2	3	4	5	
0.6	101.343	101.237	101.186	101.172	101.150	101.134	101.029
0.7	100.888	100.829	100.681	100.678	100.654	100.634	100.532
0.8	100.459	100.301	100.217	100.185	100.173	100.122	100.037
0.9	100.049	99.826	99.788	99.711	99.681	99.645	99.544
1.0	99.626	99.401	99.292	99.247	99.208	99.170	99.054
1.1	99.330	98.958	98.840	98.801	98.712	98.690	98.566

TABLE 13. V_0^{mixed} – rollup, $\delta = 2\%$, case (Ia), $g = 5.5\%$.

GMWB

$\varphi\%$	$p\%$						V_0^{static}
	0	1	2	3	4	5	
1	105.108	104.716	104.321	103.927	103.510	103.510	103.510
2	103.374	102.840	102.286	101.794	101.354	100.927	99.953
3	102.087	101.472	100.903	100.340	99.800	99.289	96.895
4	100.992	100.380	99.792	99.190	98.627	98.073	94.370
5	100.000	99.397	98.790	98.202	97.636	97.061	92.324
6	99.110	98.511	97.929	97.330	96.769	96.220	90.695

TABLE 14. V_0^{mixed} – case (Wa), $T' = 10$, $\beta = 10\%$.

the surrender penalty are insufficient to recover the cost of the guarantees, while if it is under P the proportional fee and/or the surrender penalty are too high. For instance, with a penalty of 3% the fair fee rate required for GMAB&GMDB with the mixed approach is between 8% and 9% (Table 12), against 6.229% required instead with the static approach (Table 3). Similarly, for GMIB&GMDB the fair fee is between 0.8% and 0.9% with the mixed approach (Table 13), not far from that obtained with the static approach (0.813%, Table 5). Finally, for GMWB the fair fee rate lies between 3% and 4% (Table 14) against 1.953% (Table 11).

So far we have investigated the effects of changes in the contractual parameters (roll-up rates, ratchet frequencies, guaranteed annuitisation rates, proportional fee rates and surrender penalties). Now, we perform a scenario analysis, looking at some effects produced by financial and biometric risk factors, such as interest rates, volatility of the reference fund and the force

of mortality. For these variables we have assumed square-root mean reverting processes and now choose to move their mean reverting target, keeping fixed the other parameters. Note that moving the drift of these processes allows us also, in view of market incompleteness, to catch some effects due to changes in the pricing measure. We fix a surrender penalty of 3% and, as far as financial factors are concerned, we focus on a GMWB with 10 years maturity and constant withdrawals independent of survival. In Tables 15-16 we report the contract values according to the mixed approach and, in brackets, the corresponding values under the static approach, for different levels of the instantaneous risk-free rate long term mean ζ_r , the reference fund variance long term mean ζ_K and the proportional fee rate. For biometric factors, we apply a multiplicative shift to the deterministic force of mortality $\hat{\mu}$ representing the target towards which the stochastic intensity μ reverts. More specifically, we denote by α the proportional factor applied to $\hat{\mu}$. Table 17 and Figures 2-3 help to understand the sensitivity of the time of death τ to the multiplicative shift. In Table 18 we focus on the combination of a GMIB (Ia) with 5.5% guaranteed annuitisation rate and a GMDB with 2% roll-up rate, while in Table 19 we isolate the GMIB (Ia).

GMWB					
$\varphi\%$	$\zeta_r\%$				
	0.0	1.5	3.0	4.5	6.0
1	108.828 (108.794)	106.053 (105.972)	103.927 (103.606)	101.919 (101.630)	100.517 (99.983)
2	106.735 (105.737)	104.005 (102.603)	101.794 (99.953)	100.099 (97.716)	98.845 (95.832)
3	105.179 (103.266)	102.404 (99.824)	100.340 (96.895)	98.791 (94.404)	97.637 (92.287)
4	103.962 (101.309)	101.199 (97.576)	99.190 (94.370)	97.686 (91.630)	96.543 (89.286)
5	102.954 (99.786)	100.195 (95.790)	98.202 (92.324)	96.720 (89.337)	95.571 (86.770)
6	102.110 (98.633)	99.336 (94.399)	97.330 (90.695)	95.810 (87.474)	94.677 (84.687)

TABLE 15. $V_0^{\text{mixed}}(V_0^{\text{static}})$ – case (Wa), $T' = 10$, $\beta = 10\%$, $p = 3\%$.

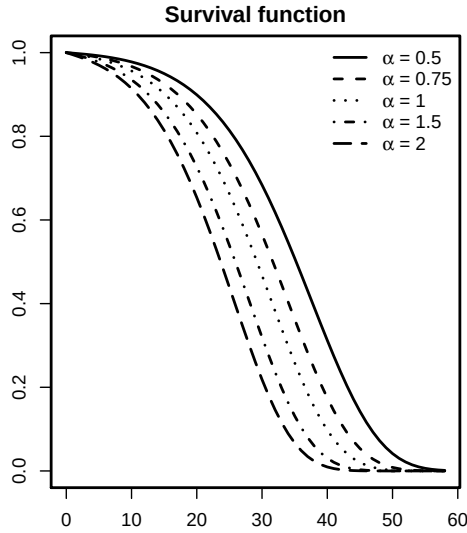
As expected, as the long term mean ζ_r increases, both static and mixed contract values decrease (see Table 15), although with different rates. The difference between V_0^{mixed} and V_0^{static} is almost always significant apart when φ and ζ_r are very low. For the long term variance ζ_K , we consider values between 0% and 8%, corresponding to volatility levels between 0% and around 28%. This parameter has a definite effect on contract values, that display a positive trend though not uniformly across different fee rates (see Table 16). Looking at Tables 18-19, we note that the multiplicative shift α has a strong impact on the contract values. In particular, it affects negatively the GMIB value (Table 19) and positively the GMDB, as expected. However, the death guarantee operates only for 5 years, when the insured is between 60 and 65 years old, while for GMIB the change in the mortality assumptions has implications for the whole life of the insured. Then, when GMIB and GMDB are combined together, the prevailing effect is that of GMIB, apart from

GMWB					
$\varphi\%$	$\zeta_K\%$				
	0	2	4	6	8
1	98.594 (97.203)	101.465 (101.223)	103.927 (103.606)	105.906 (105.671)	107.760 (107.548)
2	97.799 (93.290)	99.789 (97.606)	101.794 (99.953)	103.782 (101.946)	105.517 (103.736)
3	97.041 (90.197)	98.667 (94.637)	100.340 (96.895)	102.043 (98.794)	103.668 (100.485)
4	96.292 (88.051)	97.659 (92.259)	99.190 (94.370)	100.715 (96.149)	102.234 (97.734)
5	95.553 (86.894)	96.817 (90.411)	98.202 (92.324)	99.579 (93.955)	101.020 (95.424)
6	94.835 (86.467)	96.037 (89.015)	97.330 (90.695)	98.612 (92.163)	99.957 (93.502)

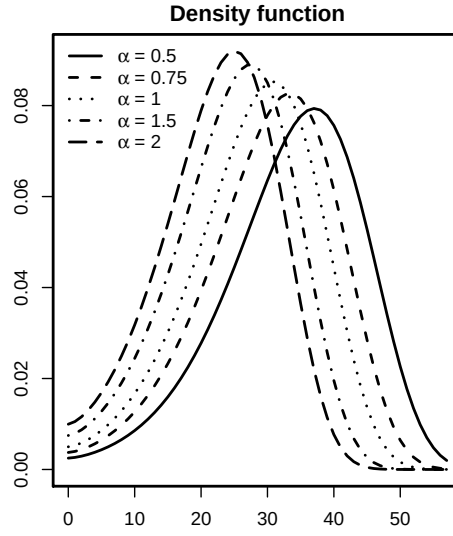
TABLE 16. V_0^{mixed} (V_0^{static}) – case (Wa), $T' = 10$, $\beta = 10\%$, $p = 3\%$.

α	$E[\tau]$	$E[\hat{\tau}]$
0.50	34.14	32.39
0.75	30.67	28.95
1.00	28.30	26.61
1.50	25.09	23.46
2.00	22.92	21.33

TABLE 17. Expected residual lifetimes.

FIGURE 2. Survival function of τ .

the case of very high mortality shocks (when α increases from 1.5 to 2), in which GMDB prevails, making the value of the combined contract slightly increasing (see Table 18). For moderate values of α (between 0.5 and 1), the differences between V_0^{mixed} and V_0^{static} are low.

FIGURE 3. Density function of τ .

GMIB & GMDB					
$\varphi\%$	α				
	0.5	0.75	1	1.5	2
0.6	113.169 (113.150)	106.223 (106.175)	101.172 (101.029)	97.586 (97.328)	97.799 (97.419)
0.7	112.604 (112.589)	105.705 (105.650)	100.678 (100.532)	97.189 (96.852)	97.234 (96.945)
0.8	112.050 (112.031)	105.187 (105.128)	100.185 (100.037)	96.858 (96.377)	96.919 (96.472)
0.9	111.492 (111.475)	104.671 (104.608)	99.711 (99.544)	96.575 (95.905)	96.712 (96.003)
1.0	110.943 (110.923)	104.154 (104.091)	99.247 (99.054)	96.463 (95.436)	96.520 (95.535)
1.1	110.398 (110.373)	103.648 (103.577)	98.801 (98.566)	96.306 (94.969)	96.347 (95.070)

TABLE 18. $V_0^{\text{mixed}}(V_0^{\text{static}})$ – rollup, $\delta = 2\%$, case (Ia), $g = 5.5\%$, $p = 3\%$.

6. CONCLUDING REMARKS

In the Nineteenth century a large variety of policies, to some extent tailored on the personal insured's needs, was customary in several European insurance markets. Later, a 'standardization' process started, with a progressive shift to a very small set of standard products, mainly consisting of the classical endowment insurance, the term insurance, the immediate life annuity, and the deferred life annuity. To some extent, an inverse process is currently developing. Indeed, many modern insurance policies are designed as 'packages', whose items may be either included or not in the product actually purchased by the client. Interesting examples are provided by insurance policies which eventually aim at constructing a post-retirement income. These products witness the shift from traditional deferred life annuities to more complex packages consisting of: one or more insurance (or

GMIB					
$\varphi\%$	α				
	0.5	0.75	1	1.5	2
0.6	113.076 (113.063)	106.090 (106.045)	100.970 (100.839)	97.420 (97.065)	97.403 (97.056)
0.7	112.513 (112.502)	105.573 (105.519)	100.462 (100.340)	97.141 (96.585)	97.132 (96.578)
0.8	111.958 (111.943)	105.054 (104.996)	99.987 (99.843)	96.902 (96.109)	96.870 (96.102)
0.9	111.404 (111.387)	104.537 (104.475)	99.481 (99.348)	96.521 (95.634)	96.569 (95.629)
1.0	110.856 (110.833)	104.032 (103.957)	99.007 (98.856)	96.448 (95.162)	96.401 (95.158)
1.1	110.306 (110.283)	103.522 (103.441)	98.644 (98.367)	96.292 (94.692)	96.158 (94.689)

TABLE 19. $V_0^{\text{mixed}} (V_0^{\text{static}})$ – case (Ia), $g = 5.5\%$, $p = 3\%$.

financial) products for the accumulation phase; an insurance (or financial) product for the decumulation phase, namely the post-retirement period.

Looking at the development of life insurance products and the management of life insurance portfolios (and pension plans) under an actuarial perspective, a progressive shift from ‘deterministic’ methods (i.e. based on the equivalence principle, and hence on expected values) to stochastic approaches is apparent.

In recent times, in particular, the awareness of financial risks and of biometric systematic risks (the aggregate longevity risk in particular) has led to the use of stochastic models also in the life insurance practice (even beyond new regulation requirements, viz. those related to the Solvency 2 project).

Variable annuities provide an interesting example of both product evaluation and need for a stochastic modelling, in particular allowing simultaneously for several risk sources, as we suggest in this paper. Moreover, under both perspectives mentioned above, further work can produce important results. From the point of view of product design, benefits (and related guarantees) could be added to the variable annuity package. For example, the uplift of the income benefits in the case of long-term-care needs could be of practical interest in several insurance markets. Under a methodological perspective, for example, single-recurrent premium arrangements should be considered, and the risk profile of the resulting accumulation structure should be assessed.

APPENDIX A. THE LEAST SQUARES MONTE CARLO ALGORITHM

We outline the algorithm employed to value a variable annuity under the mixed approach introduced in 4.4. The algorithm is based on the Least Square Monte Carlo method (henceforth LSMC), which has been proposed for the valuation of American-style contingent claims. This method combines Monte Carlo simulation with least squares regression within the dynamic programming principle used to find an optimal exercise strategy. In the life insurance field, the LSMC has recently been explored by Bacinello *et al.* (2009, 2010a). We provide now a concise, yet self contained, description

of the algorithm and refer to Bacinello *et al.* (2010a) for further details and related literature. The notation used here will be consistent with that introduced in Sections 3 and 4.

As any numerical scheme, the LSMC requires to discretize the time dimension. Without loss of generality, we take as unit of measure the discretization step and denote the time grid by $\mathbb{T} = \{0, 1, 2, \dots, N\}$, for some suitable integer N depending on the contract type. A set of state variables relevant for the valuation problem must be singled out. These variables represent financial and biometric risk factors affecting the contract under scrutiny. Typically, they include interest rates, reference fund and policy account values, mortality bases used to price life insurance contracts, and so on. We denote by X the vector of state variables. Also needed is the choice of a set of basis functions $e_1, \dots, e_k$ for some suitable integer $k > 0$. Each basis function maps the state vector X into the set of real numbers. A discussion on the choice of type and number of basis functions can be found in Moreno and Navas (2003); Gobet *et al.* (2005) contain some asymptotic results.

Step 0 (simulation): Simulate H paths of X over the time grid $\mathbb{T}$ (it may be convenient to simulate all processes over a grid finer than $\mathbb{T}$). With reference to the h th simulation ($h = 1, \dots, H$), we use the following notation:

- X_t^h vector of state variables at time $t \in \mathbb{T}$,
- M_t^h money market account at time $t \in \mathbb{T}$,
- τ^h time of death, $\tau^h \in \mathbb{T}$,
- F_t^h cash flow from the contract at time $t \in \mathbb{T}$. Depending on the riders included in the contract, it may involve: a death benefit $b_t^{D,h}$ when $t = \tau^h$; a surrender benefit $b_t^{S,h}$ if $\tau^h > t$; a survival payment $\bar{b}_t^{L,h} = B_t^{L,h} - B_{t-1}^{L,h}$ if $\tau^h > t$.

Step 1 (initialization): Set $\bar{N} = \max_{h=1, \dots, H} \tau^h$. For all $h = 1, \dots, H$,

let $F_{\tau^h}^h = b_{\tau^h}^{D,h}$ and $\lambda^h = \tau^h$.

Step 2 (dynamic programming): For $t = \bar{N} - 1, \bar{N} - 2, \dots, 1$:

- (i) Set $H_t = \{1 \leq h \leq H : \tau^h > t\}$ and, for $h \in H_t$,

$$C_t^h = \sum_{u=t+1}^{\lambda^h} F_u^h \frac{M_t^h}{M_u^h}.$$

- (ii) Regress the values $(C_t^h)_{h \in H_t}$ against $(e(X_t^h))_{h \in H_t}$, i.e. find γ_t that solves

$$\arg \min_{\gamma \in \mathbb{R}^k} \sum_{h=1}^H \left(C_t^h - \gamma \cdot e(X_t^h) \right)^2$$

(where $e = (e_1, \dots, e_k)'$ is a vector-valued function and $x \cdot y$ denotes the scalar product of vectors x and y) to obtain $\hat{C}_t^h = \gamma_t \cdot e(X_t^h)$, $h \in H_t$. If $b_t^{S,h} > \hat{C}_t^h$ then set $\lambda^h = t$ and $F_t^h = \bar{b}_t^{L,h} + b_t^{S,h}$, otherwise set $F_t^h = \bar{b}_t^{L,h}$.

Step 3 (contract values): Compute the value of the variable annuity at time 0:

$$V_0 = \frac{1}{H} \sum_{h=1}^H \sum_{t=1}^{\lambda^h} \frac{F_t^h}{M_t^h}.$$

Note that the algorithm starts at each simulated time of death and then proceeds backward, so that each process needs not be simulated over the entire time grid. Bacinello *et al.* (2010a) propose an alternative algorithm in which no time of death has to be simulated at the cost of simulating all processes over the whole time grid and replacing the money market account with a ‘mortality-adjusted’ version.

REFERENCES

- Bacinello, A., E. Biffis and P. Millossovich (2009). Pricing life insurance contracts with early exercise features. *Journal of Computational and Applied Mathematics*, vol. 233(1):27–35.
- Bacinello, A., E. Biffis and P. Millossovich (2010a). Regression-based algorithms for life insurance contracts with surrender guarantees. *Quantitative Finance*, vol. 10(9):1077–1090.
- Bacinello, A., P. Millossovich, A. Olivieri and E. Pitacco (2010b). Variable annuities: Risk identification and risk assesment. CAREFIN Research Paper no. 14/2010.
- Bauer, D., A. Kling and J. Russ (2008). A universal pricing framework for guaranteed minimum benefits in variable annuities. *Astin Bulletin*, vol. 38(2):621–651.
- Biffis, E. and P. Millossovich (2006). The fair value of guaranteed annuity options. *Scandinavian Actuarial Journal*, vol. 1:23–41.
- Bélanger, A., P. Forsyth and G. Labahn (2009). Valuing the guaranteed minimum death benefit clause with partial withdrawal. *Applied Mathematical Finance*, vol. 16(6):451–496.
- Boyle, P. and E. Schwartz (1977). Equilibrium prices of guarantees under equity-linked contracts. *The Journal of Risk and Insurance*, vol. 44(4):639–660.
- Brennan, M. and E. Schwartz (1976). The pricing of equity-linked life insurance policies with an asset value guarantee. *Journal of Financial Economics*, vol. 3(3):195–213.
- Brown, J. and J. Poterba (2006). Household ownership of variable annuities. In J. Poterba (ed.), *Tax Policy and the Economy*, vol. 20. National Bureau of Economic Research, Inc.
- Chen, Z. and P. Forsyth (2008). A numerical scheme for the impulse control formulation for pricing variable annuities with a Guaranteed Minimum Withdrawal benefit (GMWB). *Numerische Mathematik*, vol. 109(4):535–569.
- Chen, Z., K. Vetzal and P. Forsyth (2008). The effect of modelling parameters on the value of GMWB guarantees. *Insurance: Mathematics and Economics*, vol. 43(1):165–173.
- Dai, M., Y. Kwok and J. Zong (2008). Guaranteed minimum withdrawal benefit in variable annuities. *Mathematical Finance*, vol. 18(4):595–611.

- Duffie, D. (2001). *Dynamic Asset Pricing Theory*. Princeton University Press, Princeton, third edn.
- Gilbert, C., K. Ravindran and R. Reitano (2007). Results of the survey on variable annuity hedging programs for life insurance companies. Tech. rep., Society of Actuaries.
- Gobet, E., J. Lemor and X. Warin (2005). A regression-based Monte Carlo method to solve Backward Stochastic Differential Equations. *Annals of Applied Probability*, vol. 15(2):2172–2202.
- Holz, D., A. Kling and J. Russ (2007). GMWB for life - An analysis of lifelong withdrawal guarantees. Tech. rep., University of Ulm.
- Ledlie, M., D. Corry, G. Finkelstein, A. Ritchie, K. Su and D. Wilson (2008). Variable annuities. *British Actuarial Journal*, vol. 14 part II(61):327–389.
- Milevsky, M. and S. Posner (2001). The Titanic option: valuation of the guaranteed minimum death benefit in variable annuities and mutual funds. *The Journal of Risk and Insurance*, vol. 68(1):93–128.
- Milevsky, M. and T. Salisbury (2006). Financial valuation of guaranteed minimum withdrawal benefits. *Insurance: Mathematics and Economics*, vol. 38(1):21–38.
- Moreno, M. and J. F. Navas (2003). On the robustness of Least-Squares Monte-Carlo for pricing American derivatives. *Review of Derivatives Research*, vol. 6(2):107–128.
- Piscopo, G. (2009). The fair price of guaranteed lifelong withdrawal benefit option in variable annuity. *Problems and Perspectives in Management*, vol. 7:79–83.
- Pitacco, E., M. Denuit, S. Haberman and A. Olivieri (2009). *Modelling longevity dynamics for pensions and annuity business*. Oxford University Press.
- Smith, M. (1982). The life insurance policy as an options package. *The Journal of Risk and Insurance*, vol. 49(4):583–601.
- Sun, F. (2006). Pricing and risk management of variable annuities with multiple guaranteed minimum benefits. The Actuarial Practice Forum, SOA.
- Walden, M. (1985). The whole life insurance policy as an options package: An empirical investigation. *The Journal of Risk and Insurance*, vol. 52(1):44–58.